

PERSONAL INFORMATION

Akin Eker

 Bristol, UK

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 <https://akineker.github.io/>

 www.linkedin.com/in/akin-eker

EDUCATION

2021– Present

Ph.D. in Computer Science

Thesis Title: "Blockchain-of-Things: Novel System Architectures and Consensus Models as a Scalable Architecture for IoT-Driven Smart Environments"

Advisors: Prof. Theo Tryfonas and Prof. George Oikonomou

University of Bristol, UK

2018–2019

Master of Science in Computer Science

Thesis Title: "Recognising and Detecting Activities in ADL Datasets using AI"

Advisor: Dr. Leonardo Aniello

University of Southampton, UK

Taken Modules Include:

– Foundations in Machine Learning | Advanced Machine Learning | Deep Learning | Foundations of Data Science | Data Mining | Data Visualisation

2010–2015

Bachelor Degree in Computer Engineering

Dissertation: "Read a Book: A Web Platform that Allows Reading, Purchasing Books, and Note-taking and Researching on One Platform"

Advisor: Prof. M. Ali Akcayol

Gazi University, Turkey

2012–2013

Erasmus Programme in Ingeniería Informática (Computer Engineering)

University of Vigo, Spain

Awarded to students with high GPA and top scores in the English proficiency exam.

PUBLICATIONS AND ATTENDED
CONFERENCES

Conference Papers

- **A. Eker**, T. Tryfonas and G. Oikonomou, "A Conceptual Architecture for Scalable Multi-Application Support in Blockchain-based IoT Environments," *2023 IEEE 21st International Conference on Industrial Informatics (INDIN)*, Lemgo, Germany, 2023, pp. 1-6, doi: 10.1109/INDIN51400.2023.10217952.
- **A. Eker**, T. Tryfonas and G. Oikonomou, "Public Blockchain in Support of Transactive Energy Markets," *2023 IEEE 9th World Forum on Internet of Things (WF-IoT)*, Aveiro, Portugal, 2023, pp. 01-06, doi: 10.1109/WF-IoT58464.2023.10539475.
- **A. Eker**, T. Tryfonas and G. Oikonomou, "FlexBoT: A Scalable Architecture for Multi-Application Supporting BoT Environments with Application Shifting at Runtime," *2024 IFIP Networking Conference (IFIP Networking)*, Thessaloniki, Greece, 2024, pp. 186-194, doi: 10.23919/IFIPNetworking62109.2024.10619067.
- **Eker, A.**, et al., "SmartMoveChain: A Proposal for Unified Scalable BMaaS for Smart Cities." *2024 IEEE International Smart Cities Conference (ISR2)*. IEEE, 2024.
- Y. Zhang, **A. Eker**, T. Tryfonas, and N. J. Carhart, "Blockchain-based Built Environment Data Management for Modular Construction." *2024 IEEE International Smart Cities Conference (ISR2)*, IEEE, 2024.
- G. Kisa Isik, **A. Eker**, T. Tryfonas, G. Oikonomou, "Security Risk Analysis of Logistical Support Solutions for MaaS and DLT-based Mitigations." *2025 IEEE International Conference on Cyber Security and Resilience (CSR 2025)*, IEEE 2025.

Journal Papers

- **A. Eker**, T. Tryfonas and G. Oikonomou, "A Taxonomy and Comparative Analysis of Blockchain Consensus Protocols: Security, Scalability, and the Trilemma", [Under review].
- **A. Eker**, T. Tryfonas and G. Oikonomou, "Blockchain-of-Things (BoT): A Layered Architecture and Performance Evaluation for Scalable IoT Environments", [Under review].

AWARDS & CERTIFICATES

2022 Bristol Plus Award

University of Bristol, UK

An award given only 3% to the University of Bristol students annually.

2022 Innovation and Enterprise Certificate

University of Bristol, UK

2016 Turkish Ministry of Education Scholarship for Postgraduate Students

Ankara, Turkey

Given to students with high GPA and top performance in exams and interviews.

2015 Huawei Seeds for the Future Programme Trainee, Huawei Inc.

Beijing and Shenzhen, China

Selected as one of 15 students nationwide for training in Huawei's technologies and future vision in Beijing and Shenzhen.

TECHNICAL SKILLS & EXPERTISE

Programming Languages:

– Solidity | Rust | C++ | Python | JavaScript | SQL | R

Research and Theoretical Skills:

– Protocol Design | Consensus Algorithms and Blockchain Architectures | Data-driven Modelling for Decentralised Systems

Blockchain Expertise:

– Ethereum (EVM) | Layer 2 Scaling | Custom Rollup Architectures | Zero Knowledge Proofs | DeFi Data Engineering | Web3 Applications

Smart Contract Development and Tooling:

– Foundry | Hardhat | Truffle | OpenZeppelin | Chainlink | Infura | Web3.js | Ether.js | Node.js

Data and Infrastructure:

– SQL | No-SQL (MongoDB) | IPFS | Docker | AWS | Linux | Contiki-NG

Project Management:

– Git | Jira | Kanban

TEACHING APPOINTMENTS

2022 – 2025 Teaching Assistant

University of Bristol, School of Computer Science, and School of Engineering Mathematics and Technology

Assisted Modules Include (Labs):

- EMATM0066 - Financial Technology Group Project with Ethereum
- SEMTM0014 - Large Scale Data Engineering - AWS
- COMSM0089 - Introduction to Data Analytics
- SEMTM0028 - Software Development Programming and Algorithms
- SEMTM0027 - Statistical Computing and Empirical Methods
- COMS30083 - Machine Learning
- EMATM0067 - Introduction to AI and Text Analytics
- EMATM0066 - Visual Analytics
- COMSM0049 - Systems and Software Security
- COMSM0085 - Software Tools

Assisted labs, marked assessments, and supervised individual and group projects.

2016 – 2021 Research and Teaching Assistant

Gazi University, Turkey, Department of Computer Engineering

Given/Assisted Modules Include (Labs and Teaching):

- Database Management Systems with Oracle
- Data Structures with C++
- Computer Networks with C++
- Embedded Systems with C++
- Model Based Programming in Java
- Object Oriented Programming with Java
- Introduction to Artificial Intelligence with Python
- Software Engineering

Delivered labs and lectures, supervised individual and group projects, developed course materials, assessed coursework, mentored local and international students. Contributed departmental activities, including representation at academic and industry events, planning departmental tasks, and coordinating their distribution and assignment.

ACADEMIC ENGAGEMENT &
LEADERSHIP

2023 - Present

Peer Reviewer

- 2025 IEEE International Conference on Cyber Security and Resilience (CSR)
- 2024 Ad-Hoc Networks, Elsevier
- 2023 IEEE International Conference on Communications (ICC)

2022-2025

Chief Resident*University of Bristol, UK*

Facilitated student transition and wellbeing through mentoring, event coordination (open days, end of the year celebrations and etc.), team leadership, and frontline support within residences.

2021-2022

Postgraduate Hall Representative*University of Bristol, UK*

Elected by peers to represent residents, promote wellbeing, and liaise with departments.